



Innovating Public Services: Drivers and Challenges of Innovation Labs in the Co-Production of E-Government Services

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ABSTRACT

The digital transformation of the public sector towards e-government brings benefits such as greater efficiency and transparency but also faces challenges, including a lag in digital innovation and a low adoption rate compared to the private sector. This gap, exacerbated by limited IT skills and outdated processes in the public sector, undermines public trust and highlights the urgent need for governments to develop the necessary skills and technological knowledge to meet citizens' needs. Co-production in public sector innovation labs with citizen participation is an effective means of developing citizen-centered services. However, existing research is limited, and these innovation labs often struggle to engage citizens and maximize the benefits of co-production in practice. To better understand these dynamics, we conducted a study of 17 expert interviews, exploring the co-production process, the measures used, and the challenges faced. Our findings discuss the critical role of innovation labs in fostering civic engagement and managing co-production and highlight the complexity of fostering innovation in the public sector.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing; Empirical studies in collaborative and social computing;; • **Applied computing Computers in other domains** → Computing in government; E-government.

KEYWORDS

Additional Key Words and Phrases: Innovation labs, co-production, citizen engagement, case study

ACM Reference Format:

Lukas Daßler, Andreas Hein, and Helmut Krcmar. 2024. Innovating Public Services: Drivers and Challenges of Innovation Labs in the Co-Production of E-Government Services. In *25th Annual International Conference on Digital Government Research (dg.o 2024)*, June 11–14, 2024, Taipei, Taiwan. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3657054.3657138>

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dg.o 2024, June 11–14, 2024, Taipei, Taiwan

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ACM ISBN 979-8-4007-0988-3/24/06

<https://doi.org/10.1145/3657054.3657138>

1 INTRODUCTION

The integration of digital technology into government operations marks a significant shift in the interaction between governments and citizens, transitioning from traditional offline services to an increasing reliance on electronic government (e-government) services [20, 29]. E-government is referred to as “the use of information technology to enable and improve the efficiency with which government services are provided to citizens, employees, businesses and agencies” [14]. This shift offers numerous advantages, such as heightened efficiency, greater transparency, and streamlined administrative processes, enabling quicker and more convenient access to government services [13, 41, 51, 54, 55]. While e-government has made significant strides in the past two decades, the public sector lags behind industry in digital innovation. This is evident, for example, in the low usage of e-government services in OECD countries, with less than 60% of citizens using government websites [19].

This low adoption rate points to a larger issue: governments face difficulties developing the necessary skills, knowledge, and technological expertise for effective digital innovation. For example, there's an expected shortage of 816,000 professionals in the German public sector in 2030, with IT roles particularly hard to fill [6]. This shortage hampers the government's ability to offer efficient digital services centered around citizen needs. This situation, marked by reliance on outdated, paper-based processes, risks undermining citizens' trust and confidence in government effectiveness [61]. Hence, addressing this gap in skills and capabilities is crucial for governments to keep pace with digital advancements and maintain public confidence.

Research on innovation and digital transformation in the public sector often focuses on internal improvements, such as improved knowledge management [3, 53], the importance of institutional arrangements in the development and adoption of digital technologies [28, 35, 43], or the role of senior managers influencing the transformation process [4, 46, 66]. Despite alternatives such as hiring technology consultants or outsourcing tasks, most research still focuses on how the public sector develops, delivers, and innovates public services internally [7]. Confronted with the immanent challenge of e-government service development, research must consider alternative solutions to bridge the resource and skill gap.

To address these issues, scholars demand that public institutions shift to open and co-productive approaches, including citizens in the innovation process [42, 58]. Including citizens in the innovation process could mitigate resource constraints and help create citizen-centered public services [17, 60]. Innovation labs are an example of this approach, as they promote citizen engagement and collaboration between the government and the citizens [64]. Existing

research lays a solid foundation for understanding the potential of innovation labs. It includes theories and concepts [59, 62], methods [33], and empirical results [27, 39]. However, studies on innovation labs in the public sector remain scattered [7, 26], and the links to innovation in the public sector are unclear.

While integrating digital technology into government services promises significant advancements in public-citizen interactions, there remains a substantial challenge in bridging the digital innovation gap within the public sector. This challenge is compounded by a critical shortage of skilled professionals and reliance on outdated processes, raising concerns about maintaining public trust and efficiency in government operations [7, 53, 60]. Despite the potential of open and co-productive approaches, such as innovation labs for engaging citizens in the innovation process [15, 21], our theoretical understanding of their governance approaches in co-production remains underexplored. This gap in knowledge prompts the need for further investigation, leading to the research question: “*What are the drivers and challenges in the co-production process of public sector innovation labs?*”

In order to answer this question, we first present a theoretical view and then use this as a guide for the case study, which consists mainly of semi-structured expert interviews and is supplemented by secondary material [57].

The case study approach is highly suitable for investigating how public sector innovation labs foster co-production processes and innovation. We carefully selected 15 labs based on their expertise in digital transformation and their track record in projects. This selection allows us to gain comprehensive insights based on the diversity of their methods and objectives. This approach is ideal as it allows for the replication of results across different cases, highlighting both parallels and differences [67]. Despite the different objectives of the labs, a common understanding of their methods and challenges was evident. During data analysis, interviews were open-coded according to Corbin and Strauss [16], a grounded theory method that is essential for analyzing qualitative data and deriving theoretical concepts and categories [16]. The 220 interview quotes resulted in around 120 concepts, which were later classified into newly defined categories.

Our findings on innovation labs pursue the theory of co-creation of value in the public sector. It extends the theoretical understanding of the application of methods in innovation labs by showing that they use conventional and innovative experimental techniques to actively involve citizens in the idea generation and development process, thus promoting citizen-centered services while counteracting resource scarcity. Furthermore, the case study highlights two additional objectives pursued by innovation labs that go beyond pure innovation development. These include empowering citizens through a better understanding of government processes and the use of new technologies, as well as strengthening democracy by citizen participation in the development of public services and political decision-making. In practice, this research offers insights into collaboration, problem-solving, and value co-creation approaches. The results serve as a guide for projects and enable the application of identified methods tailored to specific objectives. The challenges identified in the case study enable other innovation labs to recognize and address potential problems at an early stage. In addition,

the solutions discovered in the case study provide valuable insights for the effective operation of innovation labs.

2 THEORETICAL BACKGROUND

This chapter provides the theoretical background and conceptual basis for this study. To this end, we first describe the evolution and impact of co-production in the public sector. We then give an overview of the research on innovation labs and define the relevant key aspects.

2.1 Co-Production in the Public Sector

Co-production in public administration has evolved significantly from its origins in the 1970s and 1980s, a concept introduced by Ostrom et al. [49] to understand the role of citizens in public service production [48, 50]. Initially a response to fiscal constraints, its resurgence in the 21st century is attributed to diverse governance models and global financial pressures, emphasizing collaboration with non-government actors and reinvigorating citizen roles beyond mere service users [31, 44].

The current conceptualization of co-production represents a paradigm shift in public services. Bovaird and Loeffler [9] propose a transition from “public services for the public” to “public services by the public,” highlighting the necessity of reciprocity and resource sharing for enhanced outcomes. Radnor et al. [56] and Osborne et al. [47] extend this view, emphasizing co-production as integral to public service innovation and value creation. This perspective redefines the relationship between government and citizens, underscoring the normative values of democratic governance, accountability, and social capital [23, 40, 47].

The involvement of citizens as co-producers in public services is multifaceted. They contribute as information and service providers throughout all co-production phases, transforming the traditional consumer-centric model into a collaborative, value co-creation process [30, 63]. Co-production, therefore, enhances service quality and efficiency and fosters greater citizen engagement and satisfaction with public services [2, 10, 65]. However, the concept extends beyond participation and information exchange, necessitating a reorganization of relationships and cooperation between stakeholders, including citizens, communities, government agencies, and public sector employees [43]. While economic benefits like cost savings are not the primary focus, citizen involvement often leads to more efficient service delivery and personalization of services [12, 52]. Digital technologies play a crucial role in enabling efficient and effective co-production. They allow public administrations to develop new capabilities and tasks, ensuring that employees can fully leverage the benefits of these technologies [1, 32].

In conclusion, the evolution of co-production in public services reflects a growing recognition of the importance of citizen involvement in improving service development and delivery. The approach has diversified, involving various user groups and stakeholders, each playing distinct roles in the co-production process [34, 44]. Co-production, therefore, emerges as a key strategy for realizing more democratic, efficient, and user-centric public services [11].

2.2 Innovation Labs

Innovation labs, arising from open government and evidence-based policymaking trends, are known by various names like i-lab and public policy lab [37, 64]. Despite the lack of a universal definition, Gascó [24] describes them as environments for open innovation involving collaborative research, development, and experimentation. These labs are characterized by their citizen involvement and open innovation focus innovations [18, 45], employing methods like design thinking and techniques uncommon in traditional public sector workflows.

Innovation labs employ a wide range of methods, such as designed approaches, open data analysis, evidence-based techniques, and mixed methods [37]. These labs contribute to the development of new products and services, the improvement of social standards, and the introduction of new practices. However, they face challenges like sustaining long-term civic engagement, maintaining diversity in participant groups, and overcoming the traditional public administration culture [38, 45]. Overall, innovation labs are collaborative spaces where citizens, academia, policymakers, and officials come together to foster innovation. They are characterized by diverse methods and tools and often incorporate practices from outside the public sector [33]. Current research highlights the role of innovation labs in fostering citizen-centered public services, enhancing government-citizen engagement, and incorporating various theories, methods, and empirical findings [17, 33, 60]. However, studies on innovation labs are often fragmented, and their connection to public sector innovation is unclear [7, 26].

The integration of digital technologies in government services offers significant improvements in government-citizen interactions but is hindered by a digital innovation gap, skill shortages, and outdated processes. These challenges threaten public trust and administrative efficiency [7, 53, 60]. While innovation labs have the potential to involve citizens in the innovation process, the understanding of their approaches in co-production is limited and necessitates further exploration, a gap this paper aims to address.

3 METHOD

To gain insights into the co-production approaches, methods, and processes used by public sector innovation labs and to identify the challenges in co-producing e-government services, we conducted a case study focusing on innovation labs involving citizens, comprising 15 cases and 17 interviews in total [22, 67].

For the data collection of the case study, we conducted expert interviews according to Gläser and Laudel [25]. For the selection of interview partners in this study, we initially focused on identifying experts [36] from German innovation labs in the public sector and public administrations. These individuals were either recognized for their expertise or held key leadership positions. The research began by interviewing a select group of prominent experts from German public institutions involved in digitalization and open innovation. To identify the interviewees, we used a snowball method [8]. In this approach, each expert interviewed was asked to recommend other professionals in their field who fit the criteria of the study. Interviewees were encouraged to identify peers who met the selection criteria and, if possible, make personal recommendations.

The interviews followed a semi-structured interview guideline with open questions to ensure a consistent topic and, at the same time, leave room for the specific aspects of the interview partner (IP). The interviews address two main issues: the methods and workflows regarding the co-production and innovation process as well as the obstacles and challenges to overcome during this process (the interview guideline is provided in Appendix A.1). The interviews were conducted face-to-face or via online video conference. The interview language was always based on the interviewee and was either German or English. A total of 17 semi-structured expert interviews were conducted, 15 of them with internal employees of the innovation labs, who play a key role in the co-production processes, and two with external partners on the government side and cooperation partners of an innovation lab [25]. Table 1 gives an overview of the interviewee's professional position as well as the institutional level of the innovation labs. Our selected cases represent a diverse range of innovation labs in the public sector, predominantly at the municipal level, with some at the federal level. These innovation labs differ in their specific focus areas, size, type of partners, and number of partners involved. This diversity provides a solid basis for drawing far-reaching and generalizable conclusions. For the analysis of data, we recorded and transcribed all interviews. Additionally, in order to put the interviewees' responses into context and provide a more complete overall picture, publicly available materials such as whitepapers, websites, and press releases are also included in the results [57].

In data analysis, all interviews were openly coded according to Corbin and Strauss [16]. This is a procedure from the grounded theory methodology to analyze qualitative data and to identify theoretical concepts and categories [16]. We derived about 120 concepts from more than 200 interview quotes and then grouped them by defining new categories. Our open coding scheme is illustrated by an example in Table 2.

The case study analysis is appropriate for this research as we aim to identify the co-production methods and processes used in innovation labs and describe the associated challenges from practice to extend research on innovation and co-production in the public sector [5].

Our methodology followed Yin's [67] recommendations for sampling, data collection, and analysis and was supplemented by coding procedures from Corbin and Strauss' [16] grounded theory. We used open, axial, and selective coding to develop our categories from the data and iteratively refined them until they met our initial ending criteria.

4 RESULTS

Our findings provide three key insights on the drivers and challenges of public innovation labs in the co-production process: 1) The drivers of citizen engagement, where labs actively increase participation through different recruitment approaches and the thoughtful design of the project. 2) The stages of co-production of innovation, highlighting the labs' structured approach from problem identification to implementation and emphasizing the active participation of citizens in the early stages. 3) The challenges in the co-production process, which range from recruitment difficulties to resource constraints and collaboration challenges.

Table 1: Public sector innovation labs included in the case study

ID	Position	Institutional Level	ID	Position	Institutional Level
IP1	Project Manager	Municipal	IP10	Development Coach	Municipal
IP2	Councilor	Municipal	IP11	Head of the Communication	State / Federal
IP3	Development Coach	State / Federal	IP12	Solution Architect	Municipal
IP4	Strategy Manager	Municipal	IP13	Product Owner	Municipal
IP5	Product Manager	Municipal	IP14	Agile Coach	Municipal
IP6	Lead Operations & Development	Municipal	IP15	Development Coach	Municipal
IP7	Project Manager	Municipal	IP16	Innovation Engineer	State / Federal
IP8	Project Manager	Municipal	IP17	Mayor	Municipal
IP9	Project Manager	Municipal			

Table 2: Illustration of the coding scheme

Excerpt	Concept	Categories
“For example, we have events like a developer meeting where people who like working with code can simply come together and network and exchange ideas. And we also get cool ideas about what we could potentially do with certain data sets or how artificial intelligence and art could come together in urban spaces.”	Implementation of co-produced innovations	Methods of co-producing innovations
“Sometimes there are political changes, for example, when local elections turn out differently and completely different political forces come to power, and then cooperation suddenly changes completely. This is sometimes the reason why things don’t work out so well in the end. The administration or politicians can also feel a little pressured by the demands that arise from such a competent community.”	Political challenges	Challenges

Table 3: Drivers of co-production and citizen engagement in innovation labs

Drivers	Description	Examples
Recruitment Channels	The choice of digital and analog channels to attract participants, determining the target group reached.	Digital: online ads, websites, newsletters, social media Analog: newspaper ads, public service announcements, mailing campaigns, word-of-mouth
Project Selection, Location, and Timing	The selection of community-based topics, the accessible placement of projects, and scheduling to suit diverse lifestyles are critical for enhancing participation.	Community-based topics Accessible and neutral spaces Timing events to accommodate different schedules
Information Dissemination	Disseminating information through both digital and traditional channels to improve understanding of digital and policy issues.	Interactive events and workshops, e.g., on AI or open data Exhibitions and competitions for the presentation of projects Digital platforms for the exchange of ideas and discussions

This analysis sheds light on the complex dynamics of co-production in public innovation labs by highlighting both the drivers and barriers in the process and providing a comprehensive understanding of the mechanisms and difficulties in shaping innovation in the public sector.

4.1 Drivers of Citizen Engagement in Innovation Lab

Innovation labs play a critical role in fostering civic engagement and technological innovation. To maximize their impact, innovation

labs employ various strategies to increase citizen participation in their projects. These strategies can be broadly categorized into recruitment channels, the selection of project topics, location, time and places, and dissemination of information. These drivers of citizen engagement are outlined in Table 3.

Recruitment channels. A key element in attracting participants is the choice of recruitment channels, which can be both digital and analog, as it determines the target group that is reached. Digital channels include online ads, websites, newsletters, and social media platforms such as Facebook and Instagram. These

channels offer the flexibility to run targeted campaigns to reach specific demographics. In addition, using the social media profiles of lab partners and experts can increase reach. Analog methods are still essential, especially to reach older demographics. These include advertisements in newspapers, public service announcements, and announcements by political figures. Mailing campaigns also play an important role. In addition, word-of-mouth becomes a powerful tool once a critical mass of citizens is reached. For projects requiring special skills or representation, personal invitations and organization targeting are effective, e.g., schools for student participation.

“You can reach a few people via the press and social media, but of course, depending on the topic, sometimes you have to approach people directly. Once a critical mass is reached, people will bring along their friends and colleagues who have similar interests.” (IP3)

Project selection, location, and timing. The selection, placement, and scheduling of projects are critical factors for participation. Innovation labs have more success with community-based topics as they are more accessible and impactful for the average citizen. Projects related to citizens’ daily lives or addressing urgent problems tend to attract more participants. In addition, topics that have broad appeal or that allow citizens to engage with new technologies are particularly effective in generating interest and participation.

“I think [...] if people feel a direct connection to their everyday lives, participation is easier or more successful and also [...] if you can actually witness what is happening.” (IP15)

To increase citizen engagement, the labs blend digital and physical methods. This includes apps for inquiry and discussion and contests that engage communities in local issues. These approaches are intended to give citizens’ needs and opinions greater consideration in political decision-making. Furthermore, these methods aim to inform the population about the political decision-making process and thereby legitimize it. One interviewee emphasized the importance of involving citizens in decision-making processes at an early stage and keeping them informed to counteract the accusation of pseudo-participation. To increase citizen participation on the issue of speed limits, for example, one laboratory organized a local competition between localities to motivate citizens. Participants were able to collect points by changing their driving speed or completing questionnaires, for example. The data collected was analyzed and presented at a final event where the winning municipality was announced.

The location of the event can also have a significant impact on participation. Venues such as universities can be intimidating for some, so neutral or easily accessible spaces must be chosen. For instance, one of the innovation labs uses a cargo bike as a mobile urban lab to effectively visit different locations in the city and implement projects on-site. The timing of events to suit different lifestyles and schedules is another crucial factor in reaching a wider audience.

“Discussing this is crucial to understanding the local political processes. We set up communication points in different districts where people could talk about the issue, access important information about our campaign, and

even participate in its launch. These points were mainly located at centrally located bus stops.” (IP7)

Information dissemination. Innovation labs emphasize the dissemination of information and thus facilitate the understanding of digital and political topics by demystifying advanced technologies such as artificial intelligence and the Internet of Things for citizens. Using both digital and traditional communication channels, they run interactive workshops and exhibitions on current technology trends and ensure accessibility through simple explanations and creative storytelling. The labs also offer hands-on experiences with virtual reality and 3D printing, as well as coding and digital literacy events to improve citizens’ tech skills. To ensure accessibility, the labs use simple language and creative methods such as metaphorical narratives to explain complex concepts. Digital platforms serve as online hubs for the exchange of ideas and networking to promote dialog between citizens and the administration. Virtual open citizen consultations and thematic discussion tables on these platforms further enhance effective communication.

“There was a desire for new participation formats in addition to the traditional ones, such as advisory boards, local councils, or city councils. So we decided to introduce a digital platform to make digital citizen participation a permanent feature [...] for all kinds of activities.” (IP1)

4.2 Phases in the co-production of innovation

This segment deals with creative approaches to developing new ideas, improving e-government services, and refining products. It is strongly based on the design thinking method and is divided into four phases: Identification of problems, generation of ideas, development of prototypes, and implementation of solutions. These phases are illustrated in Figure 1.

Problem Identification. Innovation labs use various methods to identify and analyze problems. Surveys are commonly used to collect quantitative data and reach citizens through various channels such as mail, email, or local events. Qualitative insights, on the other hand, are gained through interviews with users or experts. This helps to understand the application context or the intricacies of the workflow. For example, one lab included underrepresented groups such as queer and feminist organizations in its research to better understand their digital needs in urban environments. Creating personas and mapping user flows are also crucial when it comes to better understanding the different needs of different groups and identifying barriers in existing applications or processes.

Ideation. Once citizens’ problems have been identified, workshops and events are crucial for finding solutions together. This phase involves working directly with citizens to gather ideas and develop improvements. Recommendations for action are often formulated in this phase, which are later passed on to the administration for consideration.

“Our focus is intensely on idea generation because that’s where the potential lies in involving that many external individuals – it opens up a broad horizon. The aim is to gather a multitude of ideas, allowing them to be evaluated and discussed.” (IP14)

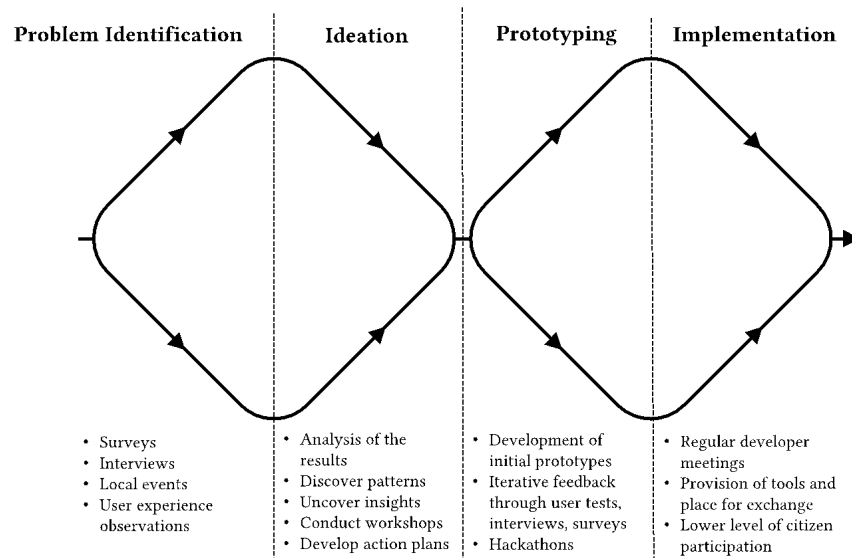


Figure 1: Phases in the co-production of innovation (own representation)

Table 4: Challenges in the co-production of innovations

Challenges	Description	Examples
Recruitment Challenges	Difficulties attracting and engaging a diverse range of citizens in innovation labs to reflect different views and backgrounds.	Difficulties in attracting a sufficient number of participants Dominance of certain demographic groups Language and cultural barriers
Resource Challenges	Financial and personnel bottlenecks are exacerbated by bureaucratic hurdles.	Underfunding and bureaucratic hurdles Misalignment of funding programs with participation processes Staff shortages
Political Challenges	Political changes and regulatory frameworks conditions that disrupt projects and limit citizen participation.	Political changes affecting projects Regulatory restrictions on citizen participation

Prototyping. In this phase, initial prototypes are created with the involvement of citizens and tested for possible improvements. The focus is on the rapid development of a minimum viable product (MVP) and subsequent iteration based on feedback. The methods range from simple sketches to mock-ups, such as a digital community center prototype. Hackathons are also a suitable method for the rapid development of prototypes. Iterative feedback obtained through surveys, interviews, or user tests is crucial for refining these prototypes.

“The core aspect of every iteration loop in our human-centered development process is that user feedback is constantly sought after each development step.” (IP16)

Implementation. This phase is less citizen-involved, mainly due to the technical nature of the programming. Nevertheless, some labs involve citizens, for example, by using open government data and encouraging them to process it for the project. Regular developer meetings allow tech enthusiasts to work together on code-based projects.

“We often simply offer a place where people who like to work with code can meet, network, and exchange ideas on a wide range of topics.” (IP4)

4.3 Challenges in the co-production of innovations

This chapter looks at the different challenges encountered in innovation labs, particularly those related to citizen engagement. These challenges are classified into four categories: recruiting citizens, managing resources, navigating political hurdles, and fostering cooperation between administration and citizens. These challenges are summarized in Table 4.

Recruitment challenges. Recruitment difficulties in innovation labs relate to the obstacles encountered when involving citizens. A major hurdle is attracting enough participants and getting them to participate in the labs’ projects. It is important, but also challenging, to recruit a broad range of citizens who reflect different views and backgrounds. Challenges include differences in education, demographics, language, and culture. Often, participation is

dominated by the same demographic groups, resulting in minorities or other community perspectives being neglected.

“We have also realized that it is quite difficult to involve society as a whole. This has to do with educational backgrounds, demographics, language, and cultural barriers that we must overcome.” (IP4)

Resource challenges. Financial constraints are common in collaborative projects with citizen labs. Underfunding by local authorities can hinder project funding, as can bureaucratic hurdles when applying for funding, especially for volunteer-run labs. Time-limited funding programs often do not fit in with the lengthy processes of citizen participation. Staff shortages, exacerbated by the pandemic, pose an additional challenge and underline the need for adequate budgets and professional facilitation to ensure the continuity of projects.

“We want to test things in practice, but you can only test them in practice if the resources are available. And if the resources are based exclusively on voluntary work, then there will be a fatigue effect among the participants at some point.” (IP6)

Political challenges. Political changes, such as the change of elected representatives, can disrupt ongoing projects and collaborations. Laboratories also face regulatory constraints that limit the participation of citizens in development processes. These changes require adaptability and continuous engagement strategies.

Collaboration Challenges. Challenges in collaboration arise from communication barriers between technology-savvy citizens and the administration, skepticism of the administration towards citizen participation, and mistrust of the public towards participation processes. Bridging the gap between citizens’ expectations and administrative processes and coordinating different agendas also present significant hurdles.

“You know, on social media, the loud voices tend to be more memorable or perhaps have a wider reach than the quiet ones. And something that really matters to me is ensuring that these quiet voices, or the ones that aren’t immediately heard, also have an impact somewhere.” (IP14)

5 DISCUSSION

This study was motivated by the intrinsic challenge of bridging the resource and skills gap and improving our understanding of the drivers and challenges in the co-production process of public sector innovation labs. To this end, we conducted an exploratory case study with expert interviews to analyze the theoretical insights on fostering civic engagement and government-citizen collaboration through innovation labs with practical insights from the public sector.

In the following, we discuss our empirical and theoretical findings on the potential of co-production processes with citizens through innovation labs. These potentials can be divided into three main categories: Citizen empowerment, which manifests itself in deeper insights into policy processes and independent use of new technologies; improving the quality of public services through

citizen participation, which leads to more effective problem identification and prioritization and thus to more diverse and relevant solutions; and strengthening democratic processes by involving citizens in design and decision-making processes, thereby enhancing trust in democratic systems and increasing acceptance of policy decisions, which in turn leads to a closer relation between the public and policy.

Finally, we reflect on the implications of our research to theory and practice and discuss the limitations of our study.

5.1 Potential of Innovation Labs in the Public Sector

This section explains the potential benefits of involving citizens in innovation lab activities. These benefits are categorized into three main areas: Empowering citizens, improving the quality of public services, and strengthening democratic processes.

Empowerment of citizens. The empowerment of citizens through their participation in innovation labs manifests itself in two different ways. Firstly, citizens gain insights into political processes, understand how parliamentary procedures work, and the responsibilities of municipal and district authorities [11, 33]. Secondly, citizens acquire the ability to use new technologies independently, as is evident in initiatives such as programming courses or open data workshops. In these workshops, for example, participants have had the opportunity to collect data independently, which increases their self-efficacy and contributes to social knowledge. This approach aims not only for the city to be the sole data authority but also for citizens to be able to measure and share data themselves. [2, 47].

Improving the quality of public services. Citizen involvement leads to the identification and prioritization of problems and projects that might otherwise be overlooked in institutionalized processes. As one interviewee pointed out, involving citizens ensures that real concerns are brought to the fore. Early citizen participation is key to obtaining a broad range of perspectives and embodies the principle of co-creation rather than presenting finished products for feedback. Involving citizens as “everyday experts” allows for a broader range of ideas and solutions that address both current and future challenges. This broader perspective helps to overcome insular thinking and ensures that different opinions are taken into account. These collaborative formats facilitated by innovation labs offer added value beyond what the administration alone can provide, not as a substitute for political bodies but as a complement to them. The inclusion of citizen feedback increases the relevance and quality of services and products, improves the interaction between the administration and the public, and reduces misunderstandings and frustration. This leads to greater satisfaction on both sides, as the expectations of the authorities are better aligned with the needs of citizens [60, 65].

Strengthening democracy. The participation of citizens in the development of public sector services and in political decision-making processes also has a democratic dimension. Citizen participation fosters a sense of agency and trust in democratic systems as individuals recognize the effectiveness of their contributions to their communities. This increased trust and the opportunity for

active participation can lead to greater acceptance of political decisions. By basing decisions on the opinions and needs of citizens, they become more legitimate and understandable. In addition, such participatory approaches can bridge the gap between the public and the political sphere, fostering closer ties and a better understanding [10, 23].

5.2 Implications

This study provides valuable insights into public sector co-production approaches, demonstrating how governments can leverage co-production for digital transformation. Its findings illuminate the drivers, potential, and challenges of innovation labs, enhancing our understanding of the role of innovation labs in government digitalization.

Theoretical Implications. This paper extends theoretical understanding in several ways. It confirms and extends the variety of methods used in innovation labs described in the literature, including both traditional and experimental approaches. The empirical results of the case study confirm and extend this diversity. While the existing literature associates these labs primarily with innovation development [24, 62], the study also shows other specific objectives that are pursued. These include participant engagement, improving digital literacy, promoting political participation, effective communication, and strengthening administrative processes. This broadens the theoretical framework for innovation labs and enables a more holistic understanding of their functions and objectives.

In researching the development of innovation, the literature emphasizes methods that focus on problem formulation and response planning rather than on the production or implementation phases [33]. The results of the case studies confirm this focus but also point to a gap in methods for involving citizens in the implementation of e-government services and products.

Practical Implications. Our research findings are valuable for both existing and emerging innovation labs. They offer a practical solution to address the lack of skilled labor and resources in public administration and the resulting low adoption of digital innovation. The empirical presentation of drivers and challenges shows how co-production in public sector innovation labs can involve citizens as active co-creators in the development of digital services, leading to needs-oriented and innovative solutions. At the same time, collaboration in formats such as workshops, hackathons, and open data initiatives promotes knowledge transfer and digital skills development among citizens and public employees. This strategy improves the quality of digital services and strengthens citizens' trust in the government, and can help reduce the IT skills shortage in the long term.

The challenges identified in the case study can guide other innovation labs to anticipate and proactively address potential problems. The solutions found in the study provide practical guidance in this area. It is crucial that innovation labs recognize challenges such as citizen recruitment and collaboration with the administration from the outset and develop strategies to mitigate them effectively.

Limitations. This study has some limitations that should be considered. Employing a case study approach based on cases exclusively in Germany, our findings are context-specific and may

not directly apply to other settings due to differences in legislation, administrative, and political-cultural aspects. Moreover, Germany's unique legal framework, which empowers local authorities with considerable autonomy, further nuances our findings. This is reflected in the varied digital participation methods and cautious implementation approaches in co-production examples.

The reliance on interviews as our main data source also introduces potential biases. The perspectives of our interviewees, particularly those deeply involved in their innovation labs, might have skewed the perceived impact and importance of various measures. Despite efforts to diversify viewpoints across different departments and hierarchical levels, the limited number of interviews may have led to certain opinions being overemphasized or overlooked.

6 CONCLUSION

In conclusion, this study comprehensively explores the integral role of innovation labs in the co-production of e-government services, particularly in the context of the public sector. It sheds light on the dynamics of citizen engagement, digital literacy enhancement, and collaborative approaches to problem-solving in government contexts. The findings from the case study and expert interviews underline the effectiveness and challenges of these labs in promoting digital transformation and civic participation. This research contributes to a deeper understanding of public sector innovation and provides practical insights for policymakers and practitioners in the field, outlining potential strategies for future development and improvement of e-government services. The study emphasizes the importance of innovation labs as catalysts for change, driving digital progress and fostering a more interactive, citizen-centric approach in public administration.

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A APPENDICES

A.1 Interview Guideline

INTRODUCTION

1. Can you describe your role and position within the organization?
2. How many years have you been a part of this organization?
3. Could you outline your key responsibilities here?

CO-PRODUCTION OF INNOVATION IN INNOVATION LABS

4. Can you walk us through a typical project lifecycle in your innovation lab?
 - 4.1. What is the average duration of projects undertaken by your lab?
 - 4.2. Can you describe the different stages a project undergoes in your lab?
5. What specific themes or types of projects does your lab prioritize?
6. Where do the requests or ideas for projects in the innovation lab typically originate?
7. What does a typical project team look like in terms of size and composition?
8. Could you explain how citizens are integrated into your innovation lab projects?
 - 8.1 How are citizens made aware of and recruited for Innovation Lab projects?
 - 8.2 What strategies are employed to encourage citizen participation?

- 8.3 What is the selection process for citizen involvement?
9. How are the results and achievements of projects communicated to stakeholders and the public?

CITIZEN ENGAGEMENT

10. At what stages can citizens participate in Innovation Lab projects, and what roles do they play?
 - 10.1 What types of engagement from citizens are most valuable to your projects?
 - 10.2 Which methods are utilized to facilitate this engagement?
 - 10.3 Can you detail the specific tasks and involvement of citizens in projects?
 - 10.4 Why were these methods of engagement chosen?
 - 10.5 Are there specific phases where citizen involvement is not utilized?
11. Is there a standardized approach to incorporating citizens into projects in your lab?
12. What resources or support does the lab provide to facilitate citizen participation?
 - 12.1 Does the lab offer training or educational programs for citizens to better engage with projects?
13. Which types of projects are most suited for citizen participation?
 - 13.1 What factors contribute to successful citizen engagement in these projects?
 - 13.2 Can you provide examples of successful citizen-involved projects?
14. How has citizen involvement impacted the development and outcomes of your projects?
15. What plans or ideas do you have for enhancing and expanding citizen participation in the future?

CHALLENGES

16. What challenges have arisen in incorporating citizens, and how have they been addressed?
 - 16.1 How has your lab attempted to overcome these challenges?
 - 16.2 What are the main barriers to citizen participation in your projects?
 - 16.3 What internal team dynamics issues arise when involving citizens?